Building Vulnerability Assessment Report

Official Technical Record — Bhuj Earthquake Event (2001)

Earthquake Event:	Bhuj, Gujarat Earthquake (26 Jan 2001 | Mw 7.7 | Depth ~17 km)
Assessment Framework:	NDMA Rapid Visual Screening (RVS) & Earthquake Field Reconnaissance
Identification Protocol:	Deterministic SHA-256 Hashing & Deduplication over Physical Structures
Report Date / Status:	October 14, 2026 | Official Engineering Assessment

1. Executive Summary & Ingestion Overview

This official report consolidates the automated post-earthquake Building Vulnerability Assessment for structures impacted by the 2001 Bhuj Earthquake. Grounded in multimodal reconnaissance data and field investigation reports (EEFIT, EERI, IITK-NICEE), physical buildings were ingested, assigned deterministic cryptographic identifiers, deduplicated, and screened against the NDMA RVS Primer.

- **Total Unique Physical Buildings Validated:** 83 (post-deduplication).
- **Deduplication Statistics:** 9 duplicate records removed via deterministic SHA-256 hashing.
- **Bhuj Vector Store Retrieval:** 1,732 report text chunks isolated specifically for the Bhuj event.
- **Primary Failure Mechanisms:** Severe prevalence of open ground storeys, floating column shear distress, torsional irregularity from rooftop swimming pools, and out-of-plane masonry wall collapses.

2. Building Breakdown by RVS Typology

All screened buildings were categorized under standard NDMA RVS typologies using rule-based classification and contextual keyword fallback search for buildings with incomplete initial metadata:

RVS Typology	Description / Structural System	Count	Share (%)
RC	Reinforced Concrete Frame (Open Storey / Infill / Shear Wall)	46	55.4%
RRSM	Random Rubble Stone Masonry (Mud / Cement Mortar)	17	20.5%
UNKNOWN	Unresolved structural classification / Insufficient cues	16	19.3%
BCBM	Burnt Clay Brick Masonry (Load-bearing walls & chimneys)	3	3.6%
DD	Dhajji Diwari (Timber / Wood Frame with Infill)	1	1.2%

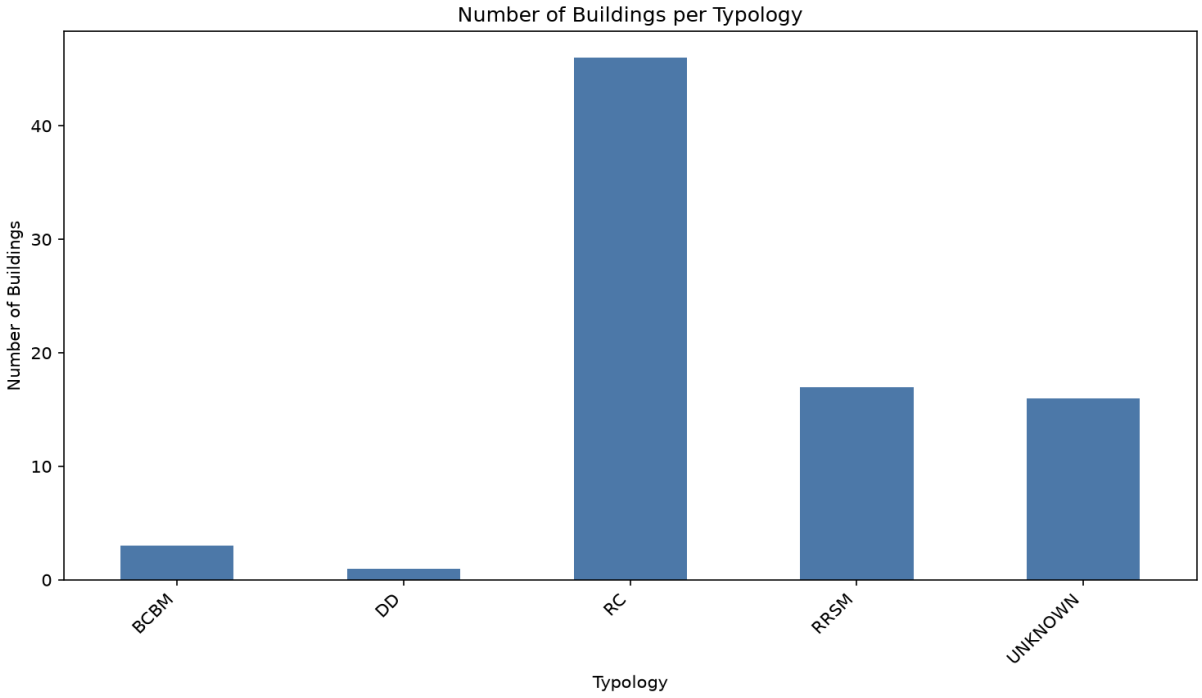


Figure 1: Building distribution across screened RVS typologies.

3. Typology-Wise Detailed Vulnerability & Frequency Analysis

Typology RC: Reinforced Concrete Frame (Open Storey / Infill / Shear Wall) (46 Buildings)

Observed NDMA RVS parameters and additional post-earthquake failure mechanisms identified across the 46 buildings of typology RC:

Source	ID	Parameter Description / Vulnerability	Freq	%
RVS	**RC_RVS_3.4**	Building has an Open Storey (at ground /other level) with slender columns.	23	50.0%
RVS	**RC_POST_3.3**	Building has an Open storey (at ground /other level) with columns having shear cracks.	23	50.0%
Additional	**ADD_RC_03**	Rooftop swimming pool causing extreme mass irregularity and asymmetric torsional collapse	9	19.6%
Additional	**ADD_GEN_01**	Common shared wall between adjacent buildings with differential response	9	19.6%
RVS	**RC_RVS_9.2**	Building has poor construction quality material.	9	19.6%
Additional	**ADD_RC_05**	Collapse of intermediate storey in commercial RC frame	8	17.4%
Additional	**ADD_RC_02**	Lack of core confinement / 90-degree ties with excessive longitudinal spacing	5	10.9%
RVS	**RC_POST_3.9**	Building has extensive X cracking or out-of-plane collapse of infills.	5	10.9%
RVS	**RC_RVS_8.2**	Building has RC walls unsymmetrical in both directions.	5	10.9%
RVS	**RC_RVS_3.6**	Building has irregular plan geometry and or plan aspect ratio > 4.0.	4	8.7%
Additional	**ADD_RC_04**	Pancake collapse of multi-storey educational frame structure	4	8.7%
RVS	**RC_RVS_8.4**	Building has unanchored water tanks on roof.	4	8.7%
RVS	**RC_POST_5.2**	Building has water tanks on roof displaced from their supports.	4	8.7%
RVS	**RC_RVS_9.1**	Building has longitudinal cracks and/or spalling in Beams and Columns.	4	8.7%
Additional	**ADD_RC_08**	Improper post-earthquake column jacketing not extended to foundations or beam joints	4	8.7%
Additional	**ADD_RC_06**	Dislodging of weakly connected overhead water tank appendages	3	6.5%
RVS	**RC_RVS_3.5**	Building has RC frame with Floating Columns.	2	4.3%
RVS	**RC_RVS_7.3**	Building has large or heavy cantilever balconies or sunshades.	2	4.3%
RVS	**RC_POST_3.4**	Building has Floating Columns with cracked supporting beams.	2	4.3%
RVS	**RC_POST_3.5**	Building has main load resisting columns with shear cracks.	2	4.3%
RVS	**RC_POST_5.1**	Building has sustained collapse of cantilevers, balconies, chimneys and parapets.	2	4.3%
Additional	**ADD_RC_10**	Insufficient connection between RC elevator core wall and adjacent framing	2	4.3%
RVS	**RC_POST_2.3**	Building is resting on liquefied soil.	2	4.3%
Additional	**ADD_RC_01**	Inadequate lap length in longitudinal column reinforcement	1	2.2%
Additional	**ADD_RC_09**	Shear cracks in cantilever stub beam supporting floating columns	1	2.2%
Additional	**ADD_RC_11**	Dislodgement of precast roof panels atop precast frame due to inadequate connections	1	2.2%
Additional	**ADD_GEN_02**	Failure of welded tie connections between prestressed concrete trusses	1	2.2%
Additional	**ADD_RC_07**	Temporary steel section or wooden post propping of damaged open ground storeys	1	2.2%

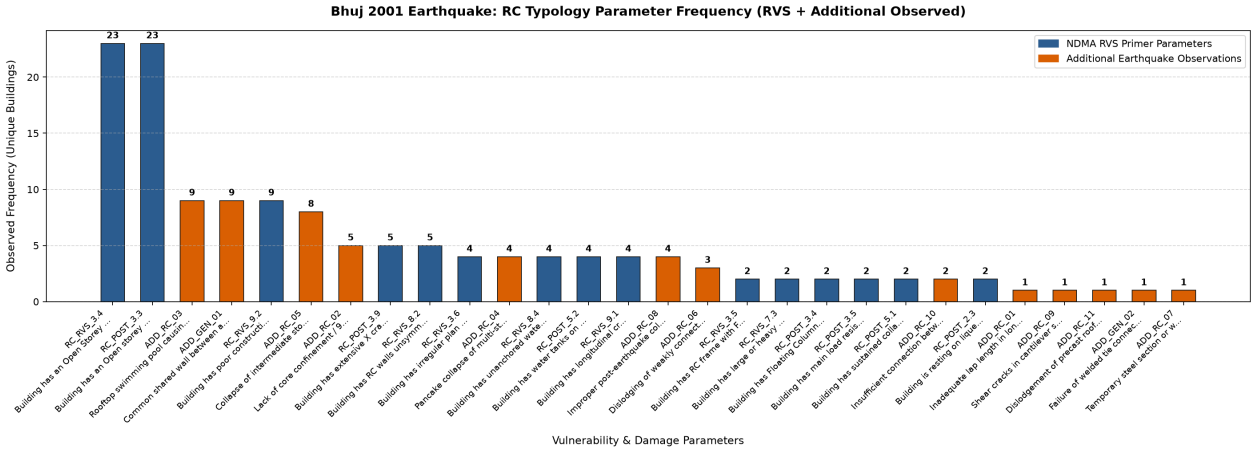


Figure: Parameter frequency distribution for typology RC.

Typology RRSM: Random Rubble Stone Masonry (Mud / Cement Mortar) (17 Buildings)

Observed NDMA RVS parameters and additional post-earthquake failure mechanisms identified across the 17 buildings of typology RRSM:

Source	ID	Parameter Description / Vulnerability	Freq	%
RVS	**RRSM_POST_5.8**	Building has floor-wall junction separated with walls out-of-plumb.	4	23.5%
Additional	**ADD_RRSM_01**	Out-of-plane collapse of stone masonry wall	3	17.6%
RVS	**RRSM_RVS_4.3**	Building has no continuous lintel band, roof band and gable band.	1	5.9%
RVS	**RRSM_RVS_4.5**	Building has masonry walls not integrated into each other at corners.	1	5.9%
RVS	**RRSM_POST_5.4**	Building has collapsed/damaged staircase/mumty or blockade of staircase.	1	5.9%
RVS	**RRSM_POST_5.7**	Building has suffered separation of wall junctions at corners.	1	5.9%
Additional	**ADD_RRSM_02**	Collapse of staircase and building corner in multi-storey stone masonry	1	5.9%
RVS	**RRSM_RVS_4.6**	Building has walls made with no or mud mortar exposed to vagaries.	1	5.9%
RVS	**RRSM_RVS_5.1**	Building has walls made with mud mortar and deteriorated and cracked significantly owing to vagaries of outside weather (especially rainwater beating).	1	5.9%

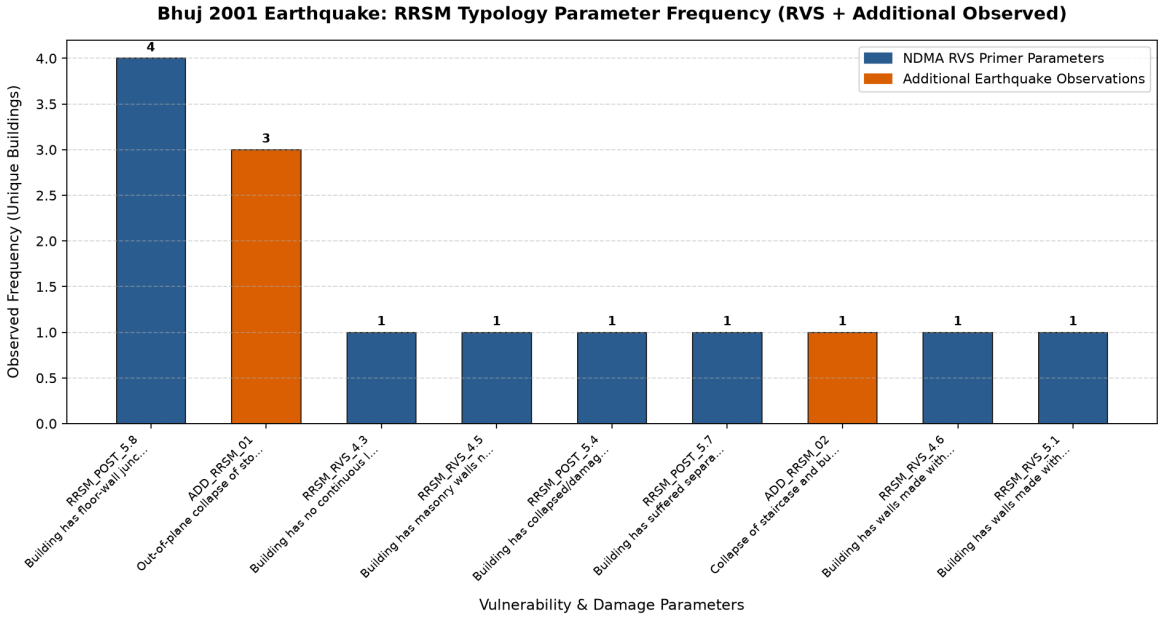


Figure: Parameter frequency distribution for typology RRSM.

Typology BCBM: Burnt Clay Brick Masonry (Load-bearing walls & chimneys) (3 Buildings)

Observed NDMA RVS parameters and additional post-earthquake failure mechanisms identified across the 3 buildings of typology BCBM:

Source	ID	Parameter Description / Vulnerability	Freq	%
RVS	**BCBM_POST_3.3**	Building has plinth masonry severely damaged.	1	33.3%
RVS	**BCBM_RVS_4.9**	Building has cracks in walls.	1	33.3%
RVS	**BCBM_POST_3.9**	Building has diagonal shear cracking in wall piers and/or spandrels.	1	33.3%
RVS	**BCBM_POST_5.1**	Building has sustained collapse of cantilevers, balconies, chimneys and parapets.	1	33.3%
Additional	**ADD_BCBM_01**	Cyclic shear cracks in industrial brick masonry chimney	1	33.3%

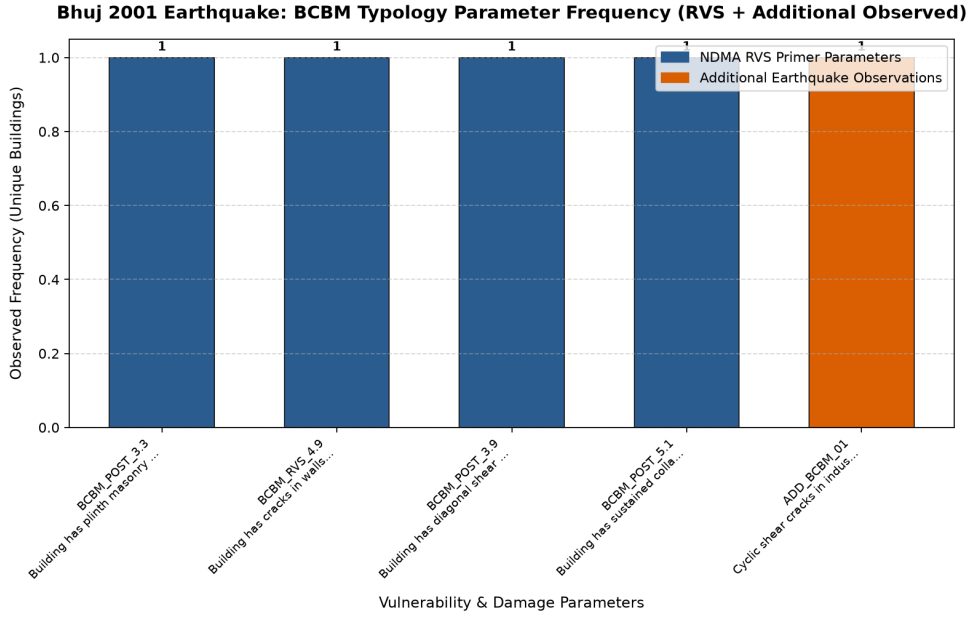


Figure: Parameter frequency distribution for typology BCBM.

4. Additional Earthquake-Observed Parameters

A critical feature of this assessment is capturing non-standard failure mechanisms documented in reconnaissance reports that exceed standard pre-earthquake RVS scoring sheets:

- **[BCBM] ADD_BCBM_01:** Cyclic shear cracks in industrial brick masonry chimney — Observed in **1** building(s) (33.3%)
- **[RC] ADD_RC_03:** Rooftop swimming pool causing extreme mass irregularity and asymmetric torsional collapse — Observed in **9** building(s) (19.6%)
- **[RC] ADD_GEN_01:** Common shared wall between adjacent buildings with differential response — Observed in **9** building(s) (19.6%)
- **[RC] ADD_RC_05:** Collapse of intermediate storey in commercial RC frame — Observed in **8** building(s) (17.4%)
- **[RC] ADD_RC_02:** Lack of core confinement / 90-degree ties with excessive longitudinal spacing — Observed in **5** building(s) (10.9%)
- **[RC] ADD_RC_04:** Pancake collapse of multi-storey educational frame structure — Observed in **4** building(s) (8.7%)
- **[RC] ADD_RC_08:** Improper post-earthquake column jacketing not extended to foundations or beam joints — Observed in **4** building(s) (8.7%)
- **[RC] ADD_RC_06:** Dislodging of weakly connected overhead water tank appendages — Observed in **3** building(s) (6.5%)
- **[RC] ADD_RC_10:** Insufficient connection between RC elevator core wall and adjacent framing — Observed in **2** building(s) (4.3%)
- **[RC] ADD_RC_01:** Inadequate lap length in longitudinal column reinforcement — Observed in **1** building(s) (2.2%)
- **[RC] ADD_RC_09:** Shear cracks in cantilever stub beam supporting floating columns — Observed in **1** building(s) (2.2%)
- **[RC] ADD_RC_11:** Dislodgement of precast roof panels atop precast frame due to inadequate connections — Observed in **1** building(s) (2.2%)
- **[RC] ADD_GEN_02:** Failure of welded tie connections between prestressed concrete trusses — Observed in **1** building(s) (2.2%)
- **[RC] ADD_RC_07:** Temporary steel section or wooden post propping of damaged open ground storeys — Observed in **1** building(s) (2.2%)
- **[RRSM] ADD_RRSM_01:** Out-of-plane collapse of stone masonry wall — Observed in **3** building(s) (17.6%)
- **[RRSM] ADD_RRSM_02:** Collapse of staircase and building corner in multi-storey stone masonry — Observed in **1** building(s) (5.9%)

Certification: *Compiled autonomously by the Seismic Damage RAG Assessment Engine. All parameters grounded directly in post-earthquake reconnaissance reports and NDMA RVS standards.*